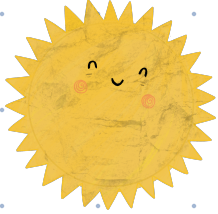


NOTES:

COMBINATORICS



"Act as if what you do
makes a difference. It does."

~ William James



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Today's content

- Addition & multiplication rule
- Permutation basics
- Combination basics & properties
- Pascal triangle
- Find n^{th} column title

Addition & Multiplication Rules

- Q1. Given 3 True/False questions, every question has to be answered. In how many ways can you answer all the questions.

$$\underline{2} * \underline{2} * \underline{2} \Rightarrow 2^3 = 8 \text{ ways}$$

Diagram showing the multiplication rule for 3 questions, each with 2 possible answers (T or F):

- Question 1: T or F
- Question 2: T or F
- Question 3: T or F

<u>F</u>	<u>F</u>	<u>F</u>
<u>F</u>	<u>F</u>	<u>T</u>
<u>F</u>	<u>T</u>	<u>F</u>
<u>F</u>	<u>T</u>	<u>T</u>
<u>T</u>	<u>F</u>	<u>F</u>
<u>T</u>	<u>F</u>	<u>T</u>
<u>T</u>	<u>T</u>	<u>F</u>
<u>T</u>	<u>T</u>	<u>T</u>

8 ways

- Q2 Given 10 girls & 7 boys. How many different pairs can we form?

Note: pair $\rightarrow$ 1 Girl + 1 Boy

Girls

Boys

G₁

B₁

G₂

B₂

1 Girl & 1 Boy

G₃

B₃

10 * 7 = 70 ways

G₄

B₄

G₅

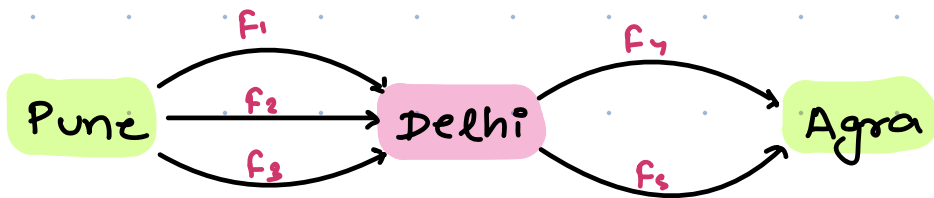
⋮

⋮

B₇

G₁₀

Eg 3



No. of ways to reach Agra from Pune via Delhi

Pune → Delhi AND Delhi → Agra

3 * 2 = 6

F₁ F₄

F₁ F₅

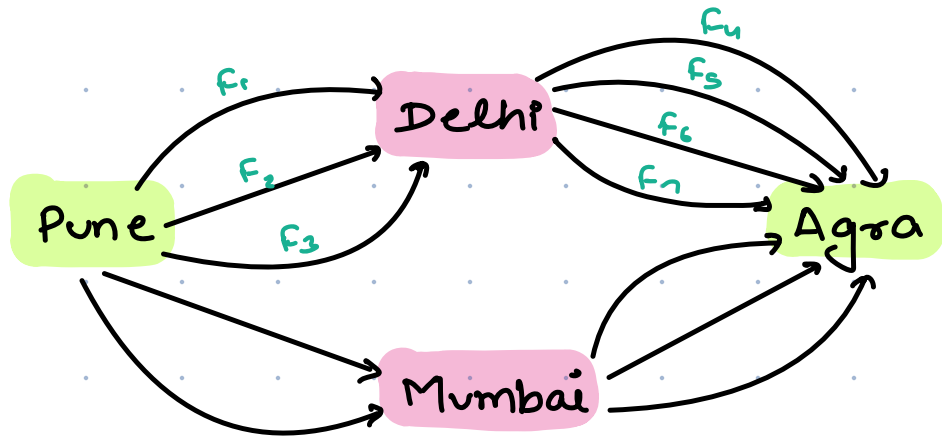
F₂ F₄

F₂ F₅

F₃ F₄

F₃ F₅

Eg 4



No. of ways to reach Agra from pune?

via Delhi

OR

via Mumbai

Pune $\rightarrow$ Delhi AND Delhi $\rightarrow$ Agra

Pune $\rightarrow$ Mumbai AND Mum $\rightarrow$ Agra

3×4

+

2×3

$12 + 6 = 18 \text{ ways}$

AND $\rightarrow \times$

OR $\rightarrow +$

* Permutation

→ Arrangement of objects

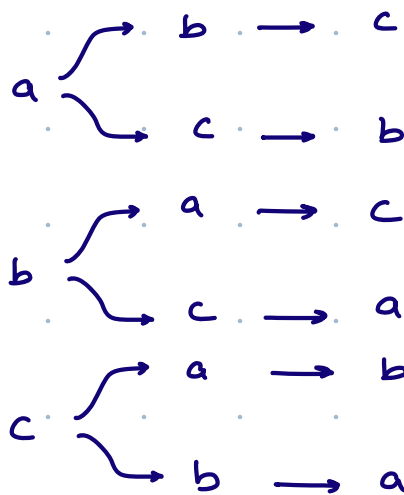
→ In permutation, order matters

$$(i, j) \neq (j, i)$$

Q Given 3 distinct characters, In how many ways can you arrange them.

str = "a b c"

$$\underline{3} * \underline{2} * \underline{1} = 3! = 6 \text{ ways}$$



* In how many ways, n distinct characters can be arranged.

$$N * (N-1) * (N-2) * \dots * 1 \Rightarrow N!$$

* Given 5 distinct characters, in how many can you arrange 2 characters.

str = "a b c d e"

$$\underline{5} * \underline{4} = 20 \text{ ways}$$

a $\rightarrow$ b, c, d, e

b $\rightarrow$ a, c, d, e

c $\rightarrow$ a, b, d, e

d $\rightarrow$ a, b, c, e

e $\rightarrow$ a, b, c, d

* Given N distinct characters, in how many can you arrange 3 characters.

$$\underline{N} * \underline{(N-1)} * \underline{(N-2)} = N * (N-1) * (N-2) \text{ ways}$$

* Given N distinct characters, in how many can you arrange r characters.

$$\Rightarrow N * (N-1) * (N-2) * \dots * (N - (r-1))$$

$$= \frac{N * (N-1) * (N-2) * \dots * (N-r+1) * (N-r) * (N-r-1) * \dots * 1}{(N-r) * (N-r-1) * \dots * 1}$$

$$\Rightarrow \frac{n!}{(n-r)!} = nP_r \Rightarrow \text{No. of ways to arrange } r \text{ places from } n \text{ distinct characters}$$

* Combination

→ No. of ways to select something.

→ Order of selection doesn't matter.

$$(i, j) = (j, i)$$

Q Given 4 players, count the no. of ways to select 3 player.

$$\{P_1, P_2, P_3, P_4\}$$

P_1, P_2, P_3

P_1, P_2, P_4

P_1, P_3, P_4

P_2, P_3, P_4

} 4 ways

Arrangements

P_1	P_2	P_3
P_1	P_3	P_2
P_2	P_1	P_3
P_2	P_3	P_1
P_3	P_1	P_2
P_3	P_2	P_1

P_1	P_2	P_4
P_1	P_4	P_2
P_2	P_1	P_4
P_2	P_4	P_1
P_4	P_1	P_2
P_4	P_2	P_1

P_1	P_3	P_4
P_1	P_4	P_3
P_3	P_1	P_4
P_3	P_4	P_1
P_4	P_1	P_3
P_4	P_3	P_1

P_2	P_3	P_4
P_2	P_4	P_3
P_3	P_2	P_4
P_3	P_4	P_2
P_4	P_2	P_3
P_4	P_3	P_2

P_1 P_2 P_3

P_1 P_2 P_4

P_1 P_3 P_4

P_2 P_3 P_4

Selection

For every selection = 6 arrangements

No. of selection $\times$ No. of arrangements = Total no. of
for each arrangements
selection

$$x \times 6 = 24$$

$$x = \frac{24}{6} = \underline{\underline{4}}$$

Q Given n distinct ele, in how many ways
can we go & select r elements.

No. of arrangements for one selection = $r!$
(pick r elements)

$$\text{Total no. of arrangements} = {}^n P_r = \frac{n!}{(n-r)!}$$

$$\text{No. of selection} * r! = \frac{n!}{(n-r)!}$$

$$\text{No. of selection} = \frac{n!}{r! (n-r)!}$$

8:00 AM $\rightarrow$ 8:10 AM

* Properties of Combination

01. No. of ways to select 0 items from N items

$${}^N C_0 = \frac{N!}{0! * (N-0)!} = \frac{\cancel{N!}}{1 * \cancel{N!}} = 1$$

02 No. of ways to select N items from N items

$${}^N C_N = \frac{N!}{N! (N-N)!} = \frac{N!}{N! \times 1} = 1$$

03. No. of ways to select r items from N items

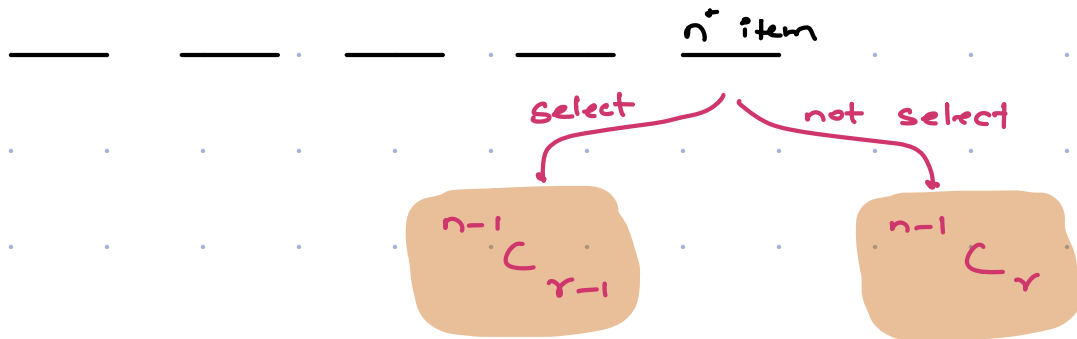
$${}^N C_r = \frac{N!}{r! (N-r)!}$$

04 No. of ways to select $(N-r)$ items from N items

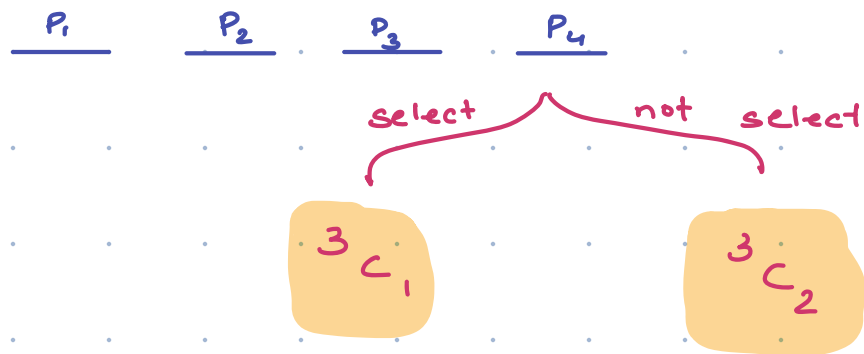
$$\begin{aligned} {}^N C_{N-r} &= \frac{N!}{(N-r)! + (N-(N-r))!} \\ &= \frac{N!}{(N-r)! \times r!} \end{aligned}$$

$${}^N C_r = {}^N C_{N-r}$$

* Given N distinct elements, select r elements



4 players & we have to select 2



$${}^n C_r = {}^{n-1} C_{r-1} + {}^{n-1} C_r$$

* Pascal Triangle

Given n , generate pascal triangle for this no.

$$\underline{\underline{N = 4}}$$

	0	1	2	3	4					
0	1						0C_0			
1	1	1					1C_0	1C_1		
2	1	2	1				2C_0	2C_1	2C_2	
3	1	3	3	1			3C_0	3C_1	3C_2	3C_3
4	1	4	6	4	1		4C_0	4C_1	4C_2	4C_3 4C_4

Brute force → For each & every position, generate a value through $N!C_r$ & print it.

$${}^N C_r = \frac{N!}{(N-r)! r!}$$

$$5! = 120$$

$$10! = 3628800$$

$$20! = 2.4 \times 10^{18}$$

factorials grow rapidly

	0	1	2	3	4
0	1				
1	1	1			
2	1	2	1		
3	1	3	3	1	
4	1	4	6	4	1

$$* \quad {}^nC_0 = 1$$

$$* \quad {}^nC_n = 1$$

$$* \quad {}^nC_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$$

$${}^4C_1 = {}^3C_0 + {}^3C_1$$

$${}^nC_r = {}^{n-1}C_{r-1} + {}^{n-1}C_r$$

$$c[i][j] = c[i-1][j-1] + c[i-1][j]$$

```

int [][] c = new int [n+1][n+1];
System.out.print("1."); // 0th row

for (i=1; i ≤ n; i++) {
    c[i][0] = 1;
    c[i][i] = 1;
    System.out.print(c[i][0]); ✓

    for (j=1; j < i; j++) {
        c[i][j] = c[i-1][j-1] + c[i-1][j];
        System.out.print(c[i][j]);
    }
    System.out.println(c[i][i]);
}

```

TC: $O(n^2)$

SC: $O(n^2)$

* Excel column Title

Find the n^{th} column title where columns are titled as:

1 2 3 ... 26 27 28 ... 52 53 54 ...
 A B C ... Z AA AB ... AZ BA BB ... ZZ AAA
 0 1 2 ... 25

$n = 3 \longrightarrow \text{Ans} = C$

$n = 27 \longrightarrow \text{Ans} = AA$

$n = 30 \longrightarrow \text{Ans} = AD$

$n = 50 \longrightarrow \text{Ans} = AX$

Series = Base = 26

$n = 52 \longrightarrow AZ$

26	52	0	$\longrightarrow X$
26	2	2	$\longrightarrow B$
	0		

- Issue $\rightarrow 0$ is not mapped with anything

26	$52 - 1$	25	→ Z
26	$1 - 1$	0	→ A
	0		

AZ

AD

* $n = 30$

29

26	$30 - 1$	3
26	$1 - 1$	0
	0	

0 3
↓ ↓
A D

* $n = 78$

26	$78 - 1 = 77$	25
26	$2 - 1 = 1$	1
	0	

1 25
↓ ↓
B Z

BZ

* String columnTitle (int n) {

ans = ""

while (n > 0) {

n = n - 1;

rem = n % 26;

n = n / 26;

char ch = (char)(rem + 'A');

ans = ch + ans;

return ans;

TC: $O(\log_{26} n)$

SC: $O(1)$

* n = 78

$$\begin{array}{r} 77 \\ - 52 \\ \hline 25 \end{array}$$

26	78 - 1 = 77	25
26	2 - 1 = 1	1
	0	

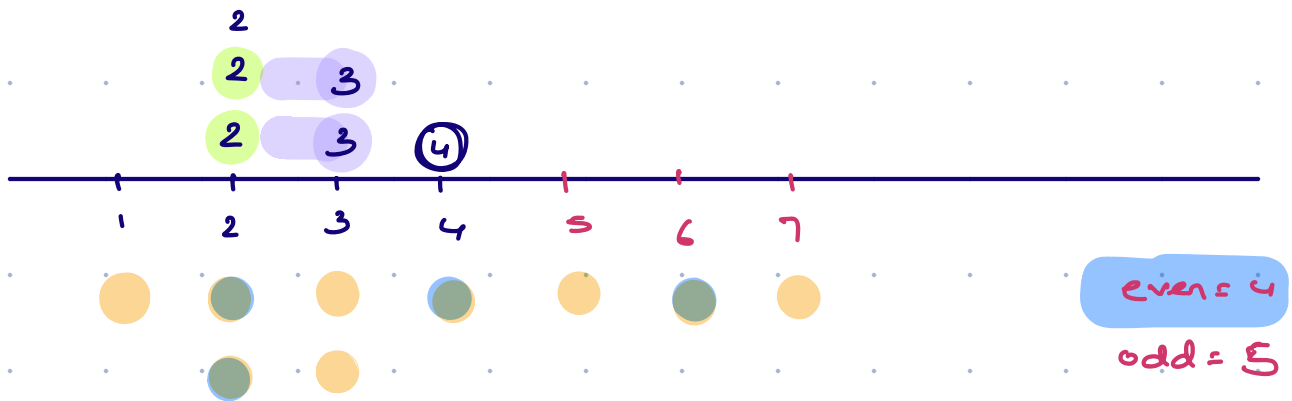
1 25
 ↓ ↓
 B Z

Doubts

Integer.parseInt("string");

$\Delta = \{ \quad , \quad , \quad , 4 \}$

Output = 2



$\text{pos}[i] + 2$ or $\text{pos}[i] - 2$ cost = 0

$\text{pos}[i] + 1$ or $\text{pos}[i] - 1$ cost = 1

$A[] = \{ 1, 2, 3, 2, 4, 6, 3, 5, 7 \}$

$\text{Min}(\text{odd}, \text{even})$